

GENERAL THETA FOUNDATIONS I: HANKEL COMPATIBILITY, DISTORTION RATES, AND FINITE-BIT MEMORY

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ABSTRACT. We place numerical orthogonal-word experiments in the finite-horizon positive Hankel realization framework and compare two previously separate complexity profiles. The optimized loss of any finite word packet, divided logarithmically by its length, is bounded by the logarithmic dilation of every compatible common polytope of the same cardinality. The proof allows arbitrary hidden stochastic states and removes initialization loss by repetition. For planar alphabets we construct a uniform table-free simulator using finite random bits and $O(\log N + \log(1/\varepsilon))$ total work bits, including parameter construction, counters and arithmetic. The explicit badly approximable families require $\Theta(\log N)$ work bits at fixed positive accuracy. The atomic label-width exponents, minimal-type metric bounds and Liouville fluctuations of the preceding development are retained with complete proofs. Atomic nonuniform width, clean register cardinality and the full finite-bit workspace are distinguished throughout. The profile comparison is one-sided; it is not a general solution of minimal positive realization or of arbitrary arithmetic width.

1. INTRODUCTION

A word response can have constant signed Hankel rank and constant nonnegative rank at every cut, yet require increasing positive state cardinality when its cut factorizations must use compatible stochastic updates. Positive realization, rather than ordinary linear realization, is therefore the relevant formalism. We study the controlled numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

for a finite orthogonal alphabet, a unit seed and one selected terminal coordinate query. Our basic resource is nonuniform clocked atomic-row positive-realization label width. The known horizon, row tables and exact real arithmetic are free in that model. A separate theorem below charges them in a uniform finite-bit command-stream model.

There are three contributions to the present treatment. First, Section 3 writes the exact finite-horizon object in the language of normalized positive Hankel factorizations and compatible future-response polytopes. This specializes the classical shift-invariant-cone principle; neither positivity nor compatibility is claimed as a new realization concept. For the planar numerical family, ordinary rank, nonnegative rank and normalized positive rank all equal three at every cut. This identifies precisely what separate factorization omits.

Second, write $\Gamma_{\mathcal{A}}(k, B)$ for optimized length- B word quantization loss and $\lambda_{\mathcal{A}}(k, a)$ for the least common dilation of a k -generator polytope containing a times the unit ball. Theorem 5.1 proves the universal comparison

$$(1.2) \quad \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B} \leq \log \lambda_{\mathcal{A}}(k, a).$$

It uses arbitrarily many independent packets to remove the initial inradius loss. Thus the paired profiles are related for every finite orthogonal alphabet, including noncommuting alphabets. Equality and optimality among all hidden realizations are not assumed.

Third, Theorem 10.3 gives a table-free finite-bit compiler for finitely many planar angles having fixed linear-space digit procedures. Its parameter search, dyadic probabilities, arithmetic scratch space, horizon counter and finite random words require $O(\log N + \log(1/\varepsilon))$ work bits. Every command is processed once; the output error is uniformly at most a prescribed $\varepsilon > 0$. For the explicit badly approximable algebraic-angle families the work-space order is exactly $\Theta(\log N)$, while the clean stochastic register retains the arithmetic label exponent. This is not an exact fair-bit implementation of arbitrary real rows, and the sharp coefficient of work space is not determined.

The packet, interval and enclosure theorems of the preceding development are reproduced with their proofs in Section 4. The packing–dilation balance there is a conditional meta-corollary, not a classification of finite alphabets. The arithmetic results below are retained: all of their analytic proofs remain in the current paper, with the branch of the finite-type minimum displayed explicitly.

Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally independent, write

$$(1.3) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

This is the ordinary dual exponent $\omega_{1,r}$ of the row vector α in [17, Definition 1], with an equivalent fixed-dimensional norm; it is not the uniform exponent $\widehat{\omega}_{1,r}$ or the simultaneous exponent $\omega_{r,1}$. The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 1.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(1.4) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C , depending on the fixed $\alpha, \delta, r, \rho, \varepsilon$, such that, for $N \geq 2$,

$$(1.5) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines. No uniformity near rational relations or in growing r is asserted.

The exponent in (1.4) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 7.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 1.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(1.6) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (1.6) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Provenance of the ingredients. The stochastic Hankel normal form and its finite separation dual are normalizations of established realization and convex-separation arguments [12, 16, 14]. Their role is to place the resource, not to claim priority for positive cones. The endpoint mechanism and arithmetic laws are inherited [7, 8, 19]. The new statements are the distortion-rate comparison and the uniform finite-bit realization with its charged resource ledger. Dirichlet approximation, continued fractions, metric tail estimates and the algebraic examples remain classical. Section 11 compares the precise objectives and norms, rather than treating adjacent theories as absent.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho < 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. In particular $W_{N,\varepsilon} \geq 2$. For fixed finite $K_\bullet$ the optimal actual error $\mathcal{E}_N(K_\bullet)$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t | S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 4.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) | w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. CONTROLLED HANKEL ARRAYS AND COMPATIBLE POSITIVE REALIZATIONS

For a horizon N , define the probability-valued prefix-suffix matrix

$$(3.1) \quad H_t[(x, u), (v, j, b)] = \frac{1 + b\rho\langle e_j, U_v U_u x \rangle}{2}, \quad |u| = t, \quad |v| = N - t.$$

The two columns indexed by $b = \pm 1$ sum to one for every fixed (v, j) . Let $\mathcal{C}_t$ be the product of these binary probability simplices. Thus a point of $\mathcal{C}_t$ is a legal collection of responses to all remaining words and queries, not a joint distribution of those answers. The restriction $L_{t,a} : \mathcal{C}_t \rightarrow \mathcal{C}_{t+1}$ reads the coordinates whose first remaining command is a .

A normalized positive factorization is $H_t = E_t B_t$, where E_t is row stochastic and every row of B_t belongs to $\mathcal{C}_t$. Its least inner dimension is denoted $\text{rank}_{\text{st}} H_t$. The usual nonnegative rank permits arbitrary nonnegative factors and need not impose these simultaneous normalizations. Consequently

$$(3.2) \quad \text{rank } H_t \leq \text{rank}_+ H_t \leq \text{rank}_{\text{st}} H_t \leq K_t$$

for every exact stochastic realization. The last inequality follows by conditioning the remaining experiment on the actual label at cut t . The factor E_t contains its state probabilities; B_t contains its complete future responses, including those of states not reached by a particular prefix.

Proposition 3.1 (Finite-horizon positive Hankel normal form). *For any finite table of binary responses, a profile $K_0, \dots, K_N$ is exactly realizable if and only if there are polytopes $P_t \subset \mathcal{C}_t$, generated by at most K_t points, such that*

$$(3.3) \quad H_0[x, \cdot] \in P_0 \quad (x \in X), \quad L_{t,a} P_t \subset P_{t+1} \quad (0 \leq t < N, a \in \mathcal{A}).$$

The held answer additionally costs two labels. The optimum error for a fixed profile is the least entrywise error from the specified table to one having such a compatible family.

Proof. Given a machine, let $b_{t,s} \in \mathcal{C}_t$ be the conditional response to every remaining word when its current label is s . Its stochastic update gives

$$L_{t,a} b_{t,s} = \sum_{s'} T_{t+1,a}(s, s') b_{t+1,s'}.$$

Taking convex hulls and using the initialization proves necessity. Conversely choose generators $b_{t,s}$, padding by repetitions when necessary. Express each $L_{t,a} b_{t,s}$ as a convex combination of the next generators, and use those coefficients as the stochastic row. Express each root row as a convex combination of the initial generators. At time N the generators themselves give the binary decoder rows. Backward induction on the remaining word proves equality of all responses. The construction is valid on every generator, not only on reachable mixtures. For approximation apply this equivalence to the actual response table of a machine, or to a candidate perturbed table; finite compactness of the stochastic parameter spaces gives attainment. $\square$

This is the finite, controlled, normalized version of the classical shift-invariant-cone realization principle, not a newly discovered characterization of positive systems. In particular, the cone generated by $(1, b)$ for $b \in P_t$ carries the positive restriction maps. Benvenuti-Farina's Theorems 2-3 give the stationary cone formulation; the multiple cuts here are essential [12]. The same compatibility issue was already explicit in the earlier lift formulation [18]. The normal form has exponentially many suffix coordinates. It is exact, but does not make arbitrary width minimization efficient or eliminate hidden directions by projecting them into the observed D -space.

Proposition 3.2 (A finite compatibility certificate). *For proposed generators B_t, B_{t+1} , put $c = L_{t,a}b_{t,s}$ and let $b'_1, \dots, b'_{K_{t+1}}$ be the next generators. The least rowwise uniform defect is*

$$(3.4) \quad \min_{\theta \in \Delta_{K_{t+1}}} \left\| c - \sum_i \theta_i b'_i \right\|_\infty = \max_{\|z\|_1 \leq 1} \left(\langle z, c \rangle - \max_i \langle z, b'_i \rangle \right).$$

It vanishes exactly when that row has a stochastic extension. If root rows are approximated within δ_0 and every cut restriction within δ_t , the resulting machine has worst-word binary-TV error at most $\delta_0 + \sum_{t=1}^N \delta_t$.

Proof. Write the infinity norm as the supremum of its pairing with the unit ℓ_1 ball. The simplex and this ball are compact and convex, so finite bilinear minimax interchanges the extrema. Minimizing the pairing over the simplex gives the maximum over its vertices with the displayed sign. For composition, restriction does not increase the infinity norm and stochastic averaging is a contraction. Induction on the suffix length therefore adds the successive errors. Each binary pair is normalized, so its total variation is exactly either coordinate error. Cancellation may reduce the true error; no equality with the local sum is asserted. $\square$

Proposition 3.3 (Constant separate ranks for the planar experiment). *For the four planar seeds, an identity command, and $0 < \rho \leq 1/10$, at every cut and every horizon,*

$$(3.5) \quad \text{rank } H_t = \text{rank}_+ H_t = \text{rank}_{\text{st}} H_t = 3.$$

Each factorization embeds in a complete machine when all other cuts are unrestricted. Nevertheless the compatible peak can grow as in Theorems 1.1 and 1.2.

Proof. The entries in (3.1) are affine functions of the two coordinates of $\rho U_u x$, so ordinary rank is at most three. Identity prefixes, identity suffixes, both coordinate queries and the four seeds give a rank-three submatrix. For the upper factorization, an equilateral triangle of circumradius 2ρ contains the disk of radius ρ . Encode $\rho U_u x$ in its three vertices v_i by barycentric coordinates. The continuation row at v_i is $(1 + b\langle e_j, U_v v_i \rangle)/2$, which is legal since $\|v_i\| = 2\rho < 1$. This is a normalized three-factor decomposition and proves the equalities. To embed it, retain the exact history until this cut, draw its factor label, retain that label and the remaining history afterwards, then use the factor decoder. Both retained histories are finite at a fixed horizon. They are not free registers in a simultaneous three-state realization. The growth assertions follow from the stated all-hidden converses, not from a claimed new separation of signed and positive realization in classical stationary theory. $\square$

4. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

4.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(4.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(4.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 4.1 (Finite-data properties). *The maxima in (4.1)–(4.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(4.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 4.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(4.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(4.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

If $\Gamma_{\mathcal{A}}(K_t, B) = 1$, the interval annihilates the conditional-centroid amplitude and is incompatible with any positive terminal calibration. This is the meaning of an infinite charge. The recurrence

is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

Theorem 4.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(4.6) \quad \mathcal{I}_N(K_{\bullet}) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(4.7) \quad \mathcal{E}_N(K_{\bullet}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_{\bullet})} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(4.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(4.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (4.5). On each selected interval put an independently sampled optimizing complete word from Proposition 4.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 4.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_{\bullet})},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (4.6) and (4.7); taking consecutive intervals proves (4.8).

For (4.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (4.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 4.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u, v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible

subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

4.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(4.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{ \lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A} \},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 4.5 (Compatible enclosure synthesis). *The finite value in (4.10) is attained and lies in $[1, a^{-1}]$. If $\rho \lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(4.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho \lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (4.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 4.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N, \varepsilon}(\mathcal{A}) \leq W_{LN, \varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 4.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x, z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N, \varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 4.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (4.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 4.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

5. A DISTORTION–DILATION COMPARISON

The lower and upper profiles defined in Section 4 do not have identical quantifiers. They nevertheless obey a universal inequality. Define the optimized per-command distortion rate

$$(5.1) \quad \mathfrak{r}_{\mathcal{A}}(k) = \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B}.$$

All logarithms in this paper are natural. The value $+\infty$ has its usual extended meaning.

Theorem 5.1 (Distortion bounds every common enclosure). *For every finite orthogonal alphabet and every feasible k, a in (4.10),*

$$(5.2) \quad \mathfrak{r}_{\mathcal{A}}(k) \leq \log \lambda_{\mathcal{A}}(k, a).$$

In particular, every finite-word certificate gives the lower enclosure bound $\lambda_{\mathcal{A}}(k, a) \geq (1 - \Gamma_{\mathcal{A}}(k, B))^{-1/B}$. The assertion does not require commuting commands, a spectral gap, or any Diophantine hypothesis.

Proof. Take any feasible polytope $P = \text{conv}(v_1, \dots, v_k)$, contained in the unit ball and containing a times that ball, with common dilation λ . Use the stochastic rows representing $\lambda^{-1}U_b v_s$ by the same vertices. Fix a unit seed u and initialize the mean at au . This construction makes sense at every length, without imposing a terminal decoder. For each complete word w of length n it satisfies

$$\mathbb{E}[v_{S_n} \mid w] = a\lambda^{-n}U_w u.$$

Let $Y_n = U_w u$ and $q_n = \mathbb{E}\|\mathbb{E}[Y_n \mid S_n]\|$. Pairing with the bounded vector v_{S_n} gives, under every word law,

$$(5.3) \quad q_n \geq \mathbb{E}\langle Y_n, v_{S_n} \rangle = a\lambda^{-n}.$$

Fix B and independently repeat an optimizing length- B word law m times. The packet lemma applies at the m endpoints, each of which has at most k labels. Since $q_0 = 1$,

$$a\lambda^{-mB} \leq q_{mB} \leq (1 - \Gamma_{\mathcal{A}}(k, B))^m.$$

Taking m th roots and letting $m \rightarrow \infty$ removes the inradius factor. Hence $\lambda^{-B} \leq 1 - \Gamma_{\mathcal{A}}(k, B)$. Take the supremum over B and then the minimum over enclosures. In particular, $\Gamma = 1$ is incompatible with a finite feasible enclosure at this k , so no undefined logarithmic operation was used. $\square$

The removal of the initial inradius loss by repetition is important: a single-horizon application would leave an additive $\log(1/a)$. Theorem 5.1 is a comparison, not a duality asserting equality or a characterization of arbitrary hidden width. The Hankel normal form in Proposition 3.1 permits general future-response polytopes, whereas (4.10) uses one common polytope in the observed vector space.

Corollary 5.2 (Nonuniform enclosure chains). *Suppose $P_t \subset \mathbb{B}^D$ has at most K_t generators, $a\mathbb{B}^D \subset P_0$, and $U_b P_{t-1} \subset \lambda_t P_t$ for every command. Then*

$$\mathcal{I}_N(K_\bullet) \leq \log(1/a) + \sum_{t=1}^N \log \lambda_t.$$

Proof. Use the cut-dependent convex rows and the initialization with mean au . The same induction gives mean $a(\prod_t \lambda_t)^{-1} U_w u$. Choose the independent packet law attaining the interval recurrence. The pairing argument bounds the final amplitude below by this factor, while the packet theorem bounds it above by $\exp(-\mathcal{I}_N(K_\bullet))$. Taking logarithms proves the claim. $\square$

6. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (4.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(6.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

Theorem 6.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(6.2) \quad \Gamma_A(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(6.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P}, A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P}, A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. The decoder centers are on the unit circle; maximizing their inner product is therefore exactly nearest chordal-distance assignment.

For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (6.1). Thus the maximum over centers equals $F_k(p)$. This proves (6.2); introducing the norm epigraphs proves (6.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs

$O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (6.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 4.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 6.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (6.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (4.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 6.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_{\mathcal{A}}(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (6.2), not necessarily its exact value. $\square$

7. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(7.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 7.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L - 1)$. Then*

$$(7.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(7.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(7.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L - 1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L - 1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 6.3 and (4.8), with $-\log(1-x) \geq x$, give (7.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(7.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2 \eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (7.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 4.5 with $a = 1/2$ applies: $\rho \Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1} R_{2\pi\alpha_i} v_s$ by the vertices for each row, and decode by $2\rho \Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r+1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (7.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (7.2) can be weak when long word packets have clusters; the optimized profile of Section 6 is available without that separation relaxation.

Lemma 7.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(7.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(7.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq r q^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1} q^{(\tau+1)/r}.$$

Rearranging yields (7.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 7.3 (Power bounds at finite Diophantine type). *Under (7.6), for constants depending on the fixed data,*

$$(7.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (7.2), $N < B + b^{-2} B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 7.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2} N Q^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (7.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (7.4) proves the first. $\square$

Corollary 7.4 (The retained badly approximable law). *If (7.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (7.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

8. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (1.4) in Theorem 1.1. If $\omega^*(\alpha) = r$, then for every $h > 0$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 7.3 with $\tau = r + h$. It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \min \left\{ \frac{1}{2}, \frac{r(1+h)}{2r+1+h} \right\}.$$

For $0 < h < 1/(2r-1)$ the second branch is smaller. Let $h \downarrow 0$. This proves (1.4). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 8.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(8.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^(\alpha) = r$.*

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersect the full-measure sets for rational $s > 1$. For a real $s > 1$, choose a rational $1 < s_0 < s$; its stronger bound implies the bound with s , after a harmless change of constant. This gives the assertion on one full-measure set.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M+1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M+1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 8.2 (Quantitative bounds under logarithmic separation). *Suppose (8.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(8.2) \quad cN^{r/(2r+1)}(\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N+2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (7.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2+B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2+K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2+K) \leq \log(2+N)$, and rearrangement proves the first inequality of (8.2).

For the upper inequality repeat the residue argument of Lemma 7.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2+H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(8.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1} (\log(2+H))^s \leq C_r q^{1+1/r} (\log(Q+2))^s$, and raising the rearranged inequality to the r th power gives (8.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (8.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q+2) \leq C_3 \log(N+2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 7.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 1.1. Apply Lemma 8.1 with $s = 1 + \delta$ and then Proposition 8.2. This yields (1.5) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

8.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 8.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (1.5), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. The peak, including the held answer, is $\max(2, hq)$, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

9. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 9.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(9.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n\alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$ the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 9.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(9.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 9.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 6.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (7.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 9.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (7.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 1.2. Equation (9.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 9.3 gives the zero lower limit for Liouville angles, and (7.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K-1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K-1$ imply $\lfloor N/(2K-1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N, \varepsilon} \rightarrow \infty$. $\square$

Corollary 9.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 9.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (7.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

10. FINITE RANDOM BITS AND UNIFORM SPACE

Exact real rows and finite-bit algorithms are different resources. We first give a finite perturbation statement and then remove the row table from a concrete uniform construction. Only this section uses a computational model. It does not change the exact atomic-row conclusions proved elsewhere.

Lemma 10.1 (Dyadic perturbation of a stochastic transducer). *Suppose initialization and every command row have at most s nonzero entries. They can be replaced by denominator- 2^b rows, on the same label sets, with row-TV error at most $(s-1)2^{-b}$. Rounding the terminal binary probabilities adds at most 2^{-b} . Consequently every numerical word changes by at most $((N+1)(s-1)+1)2^{-b}$ in binary total variation. Each rounded row can be sampled with exactly b independent fair bits.*

Proof. Order the s successors, round the first $s-1$ probabilities down to multiples of 2^{-b} , and give the remaining mass to the last successor. The total removed mass is at most $(s-1)2^{-b}$, which is exactly the row total variation. Couple the initializations and then the command updates, or telescope products using TV contraction, to sum their worst-row errors. A terminal Bernoulli parameter can be rounded down with error at most 2^{-b} . A uniform integer in $\{0, \dots, 2^b - 1\}$ selects one of the cumulative dyadic intervals. This uses a fixed number of bits, with no retained random tape or rejection-time side channel. $\square$

For a table implementation the storage is $O(|\mathcal{A}|Ks(\log K + b))$ bits for common rows of width K , in addition to initialization and decoder data. Cut-dependent rows can require a factor N more storage. Lemma 10.1 is the elementary stochastic-rounding principle; it is not claimed as a new finite-bit simulation method. The next construction uses no such table.

Definition 10.2 (Uniform command-stream model). A fixed finite program receives N in binary, one of the four planar seeds, exactly N one-way command symbols, and one final coordinate query. It has a work tape and fresh independent fair bits. The command input cannot be reread. All counters, arithmetic registers, stored parameters, random words and output storage count toward work space. An external source waits while a command is processed; waiting time is not an input or a persistent storage channel. The finite control is fixed independently of N . The program has no horizon-dependent advice or transition table.

An angle has an admissible digit procedure if a fixed program computes a dyadic approximation within 2^{-p} in polynomial time in p and $O(p)$ work bits. The same fixed r is used throughout. This explicit hypothesis excludes a free oracle for an arbitrary noncomputable real. Algebraic angles, in particular the examples below, meet it by rational bisection with fixed-degree integer arithmetic.

Theorem 10.3 (A table-free finite-bit compiler). *Fix r angles with admissible digit procedures, rational $0 < \rho \leq 1/10$, and rational $0 < \varepsilon < 1$. There is a uniform program in Definition 10.2 whose every binary output row has TV error at most ε from the rotation experiment. It uses*

$$(10.1) \quad O_{r,\alpha,\rho}(\log(N+2) + \log(1/\varepsilon)) \text{ work bits}, \quad (N+2)b \text{ fair bits},$$

where $b = \lceil \log_2(16(N+2)/\varepsilon) \rceil$. Its total running time is polynomial in $N + \log(1/\varepsilon)$ for fixed digit procedures. Its clean stochastic register has $q+1$ labels, including one central label, with $q = O(\sqrt{N})$.

If a denominator $q_0 \geq 4$ satisfies $N\eta_\alpha(q_0) \leq q_0^2/2000$, the search may choose $q \leq q_0$. Consequently the clean-register upper exponents and logarithmic bounds of Theorem 1.1 hold for this compiler as well, with changed constants. These clean labels do not include its scratch space or deterministic parameter registers; the full work bound is (10.1).

Proof. We give the parameter search, real reference rows, dyadic rounding, and finite-space evaluation separately.

Parameter search. Put $p = \lceil \log_2(16000(N+2)/\varepsilon) \rceil$ and obtain dyadic $\tilde{\alpha}_i$ with error at most 2^{-p} . For $q = 4, 5, \dots$ compute, by integer arithmetic,

$$e(q) = \max_i \|q\tilde{\alpha}_i\|_{\mathbb{R}/\mathbb{Z}} + q2^{-p}.$$

Stop at the first q for which $Ne(q) \leq q^2/1000$. The test is rational and certifies both $\eta_\alpha(q)$ and $\eta_{\tilde{\alpha}}(q)$ by the Lipschitz property of distance to $\mathbb{Z}$. It succeeds by $q = \max(4, \lceil \sqrt{2000(N+2)} \rceil)$:

$\|q\tilde{\alpha}_i\| \leq 1/2$ and the added term is smaller than one at this bound. If q_0 satisfies the stronger displayed condition, then $e(q_0) \leq \eta_\alpha(q_0) + 2q_02^{-p}$ and $2Nq_02^{-p} \leq q_0^2/2000$. Hence that candidate succeeds. No exact comparison of an irrational angle with a rational sector boundary is performed.

Reference rows with slack. Use the regular q -gon $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and one additional label $*$ representing zero. Set $\Lambda = 1 + 1/(4N)$. For $N \geq 1$ the dilation estimate (7.5) and the search condition give

$$\log \Lambda \geq 1/(5N) > 4\pi^2\eta_\alpha(q)/q^2.$$

Thus $\Lambda^{-1}R_{2\pi\tilde{\alpha}_i}v_j$ lies strictly inside the polygon. The case $N = 0$ is handled by direct rounding of the seed response and a fixed amount of space. Write $q\tilde{\alpha}_i = n_i + t_i$, with integer n_i and $0 \leq t_i < 1$, and put $h = 2\pi/q$. The command row has successors $j + n_i$, $j + n_i + 1$ modulo q , and $*$, with respective weights

$$(10.2) \quad A_i = \frac{\sin((1 - t_i)h)}{\Lambda \sin h}, \quad B_i = \frac{\sin(t_i h)}{\Lambda \sin h}, \quad 1 - A_i - B_i.$$

They are nonnegative by the dilation bound, and their vector mean is the required contracted rotation. The row at $*$ stays there. Initialize with vector mean $x/2$ using the same chord formula with radial factor $1/2$ and the exact seed angle $0, 1/4, 1/2, 3/4$. Decode with mean $2\rho\Lambda^N\langle e_j, v_s \rangle$, or zero at $*$. Since $\Lambda^N \leq e^{1/4}$, all decoder means are strictly inside $[-1, 1]$. The usual mean induction realizes every word for $\tilde{\alpha}$ exactly. These are reference real rows only.

Certified finite rows. Compute outward intervals for the first two weights in each row, each of width at most 2^{-b} . Clip their lower endpoints at zero and round down to denominator 2^b ; assign the residue to $*$. Relative to the reference row, at most $4 \cdot 2^{-b}$ mass is moved. Evaluate and round each terminal Bernoulli parameter within $2 \cdot 2^{-b}$. TV contraction therefore bounds initialization, N updates and final decoding by $(4N + 6)2^{-b}$. Replacing the true angles by their dyadic approximants changes a product rotation by at most $2\pi N2^{-p}$ in operator norm, and changes a target binary row by at most $\pi\rho N2^{-p}$. The sum is less than ε with the chosen p, b . Use exactly b fair bits at each draw, including dummy bits when the central row is deterministic. The random-bit count is $(N + 2)b$.

Work and uniform evaluation. Only q , the current vertex or central flag, the horizon and command counter, the r dyadic angles, and $O(r)$ offsets and dyadic weights are retained. The latter specify circulant rows, not a q -row table. The denominator search uses integers of $O(p + \log q)$ bits. Elementary sine and cosine on bounded intervals can be evaluated by fixed-point Taylor recurrences with outward rounding; π can be evaluated by Machin's arctangent identity. There are $O(b + \log q)$ terms and only a fixed number of live accumulators. Since $\sin(2\pi/q) \geq 4/q$ for $q \geq 4$, division loses only $O(\log q)$ bits. Repeated squaring evaluates Λ^N with $O(\log N)$ stages, all intermediate true values between 1 and $e^{1/4}$. An additional $O(\log(N + 2) + \log(b + 2))$ guard bits bounds accumulated rounding errors. Interval precision can instead be doubled until the stated widths are certified; these error estimates bound the final precision by the same order. No integer of unbounded exact denominator is stored. Every live register has $O(b + p + \log N)$ bits, and the number of registers is fixed for fixed r . This proves the work bound and a polynomial time bound, including parameter search. Finally the finite arithmetic upper proofs remain valid with $1/100$ replaced by $1/2000$, since only fixed constants change. Their candidate denominators therefore bound the search. $\square$

The proof describes a fixed-point implementation, not the numerical behavior of a floating-point library. The accompanying integer-interval reference code certifies each rounded row and uses fresh bounded-length random integers; its Python object overhead is not identified with the Turing work-space constant.

Corollary 10.4 (Sharp work-space order for explicit alphabets). *Let $\alpha_i = 2^{i/(r+1)}$, $1 \leq i \leq r$, and fix rational $0 < \varepsilon < \rho/(2\sqrt{2})$ and $0 < \rho \leq 1/10$. In Definition 10.2, numerical simulation requires and suffices with $\Theta(\log N)$ work bits. More precisely every fixed program using $s(N)$ work bits obeys*

$$(10.3) \quad s(N) \geq \frac{r}{2r+1} \log_2 N - O(\log \log(N+2)).$$

The compiler's clean register has $O(N^{r/(2r+1)})$ labels. This is not a claim that all its internal fair-bit states retain that label exponent or that the leading coefficient of work space is optimal.

Proof. Put $\theta = 2^{1/(r+1)}$. For a nonzero integer vector m with $H = \|m\|_1 \geq 1$, choose the integer p nearest to $-\sum_i m_i \theta^i$. The algebraic integer $P(\theta) = p + \sum_i m_i \theta^i$ is nonzero: the minimal polynomial $t^{r+1} - 2$ is Eisenstein and has degree $r+1$. Each of its other r conjugates has modulus at most $(9/2)H$, since $|p| \leq 2H + 1/2$ and $|\theta^i| < 2$. Its nonzero integer norm gives

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq (2/9)^r H^{-r}.$$

This classical algebraic example is credited in [4]; it is not new arithmetic. The inherited badly approximable theorem, Corollary 7.4, now supplies the all-hidden lower bound and the matching candidate denominators for the compiler. Bisection for θ and fixed-degree powers meet the digit hypothesis. A fixed program with $s(N)$ work bits has at most $C(s(N) + 1)^{C2^{s(N)}}$ command-boundary configurations. Its private randomness since the last boundary is integrated into the next stochastic row, so it induces a legal atomic realization of at most that width. Apply the lower bound $W_{N,\varepsilon} \geq cN^{r/(2r+1)}$ and take logarithms. The polynomial factor accounts for the fixed number of work-tape head positions; it contributes $O(\log \log N)$ unless the displayed lower bound is already automatic. Theorem 10.3 supplies the upper work bound. $\square$

11. REALIZATION THEORY, ARITHMETIC AND COMPUTATIONAL SEMANTICS

Positive order and cone invariance. In the classical stationary problem one seeks a fixed nonnegative state-space realization of a transfer function. Minimal signed order need not equal minimal positive order. Benvenuti–Farina's Theorems 2–3 formulate positive existence by an invariant polyhedral cone, either in a minimal linear realization or in a response-sequence space [12]; the 2022 survey concerns positive minimality [13]. Our enclosure construction is such a positive-cone construction after homogenization. The competing machines are different: their rows may vary with the cut and their entire realization may be redesigned for N . Proposition 3.1 makes the corresponding finite controlled normal form explicit. Classical stationary order cannot be substituted for its least simultaneous profile.

Hankel rank and signed weighted automata. For fixed seed and query the mean series is $w \mapsto \rho\langle e_j, U_w x \rangle$, with a signed linear representation of dimension at most D . Adding the constant probability offset requires at most $D + 1$ dimensions. Balle–Panangaden–Precup's Theorem 2 identifies minimal weighted-automaton dimension with infinite Hankel rank [14]. That is the signed, all-length invariant, not stochastic label cardinality. Their Theorem 11 bounds an ℓ^2 approximation of absolutely summable rational series. Here the unweighted all-word probability series is not summable, and the objective is the maximum numerical error on a fixed length. A small weighted mean-square error under a chosen word law does not control rare-word conditional error. For an explicitly unweighted finite table, Frobenius error does bound entrywise error, but its optimizer still need not be stochastic or compatible. Equation (3.5) exhibits the precise failure of separate ranks to detect the present growing resource.

Stochastic and hidden Markov realization. Finesso–Grassi–Spreij use pseudo-Hankel nonnegative factorizations and an informational-divergence criterion to approximate a stationary output process at prescribed hidden order [15, Sections III–IV]. Their second realization step is itself evidence that a static factorization is not the complete dynamic specification. Our alphabet is externally controlled rather than generated as an observation process. Seed resets and selected query columns are separately specified; worst-word conditional probabilities replace a divergence between output laws. These are explicit changes of datum and objective, not a claim that Hankel positivity was absent from HMM theory.

Ohta’s reachable and null-space reductions, Theorems 6–9 and Remark 10, identify tensor and generalized-Hankel dimensions in the effective linear space of an HMM [16]. We use the same need to distinguish observable rank from hidden state count, but do not infer a positive stochastic basis from a signed quotient. His Section 4.4 also discusses rotational examples descended from earlier HMM realization work. Rotation-based rank/positive-order separations are therefore not claimed new here. Our quantitative theorems concern all horizon-specific stochastic rows and preserve bounded numerical responses on every controlled word.

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 7.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 8.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 4 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 4.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 4.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also

established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. The exact arithmetic upper rows are atomic real kernels. Section 10 gives a separate finite-bit approximation theorem; it does not represent those exact rows by a finite number of fair bits. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Exact finite widths, the Liouville upper logarithmic limit, optimal enclosure shapes and equality in the distortion–dilation comparison remain distinct quantitative problems. The new profile comparison uses no external spectral gap. For fixed computable planar angles, storing all command counts already gives a classical $O(\log N)$ finite-accuracy work-space upper bound. We do not claim novelty for that upper order alone. The compiler proved here additionally maintains an arithmetic-scale clean stochastic register, generates circulant dyadic rows without a large advice table, and explicitly prices every arithmetic and random-bit register. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

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